Lecture 25-26
Bias, Variance & dimension

- build models on the training dataset & evaluate a suitable model using data not used in fitting
  Training → fit model params
  Validation → tune HPT
  Test → evaluation
- Global model uses a shared mathematical structure across entire input space. Local model uses info. primarily from the neighbourhood of the particular input being predicted.
- Least sq. → 1 shape everywhere

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x + \cdots + \hat{\beta}_p x_p$$

$$\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{N} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

↳ global as every pt. influences the estimated coefficients there. the mini. problem. The error produced here is bias.

- KNN → let the neighbours decide.

$$\hat{f}(x) = \frac{1}{K} \sum_{i \in N_K(x)} y_i$$

when we need a predic" at x, we take its nearest neighbours & avg. their responses.

$$f(x) = E(Y \mid X = x)$$

- $$MSE_{train} = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{f}(x_i) \right)^2$$

$$MSE_{test} = \frac{1}{m} \sum_{j=1}^{m} \left(\bar{y}_j^{test} - \hat{f}(\bar{x}_j^{test}) \right)^2$$

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- Overfitting → low train error, highest error
  Underfitting → high train error

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	least Sq.	KNN
Geometry	Rigid	Flexible
ability to learn	limited	high
bias	↑	↓
vari.	↓	↑

- Bias → diff. btw. avg. fitted predic^n & true func^n

$$\text{Bias}(\hat{f}(x_0)) = E_D[\hat{f}(x_0)] - f(x_0)$$

- Vari → how much predic^n Δs when we train the same algo. on diff. random datasets

$$\text{Var}(\hat{f}(x_0)) = E_D\left[\left(\hat{f}(x_0) - E_D\hat{f}(x_0)\right)^2\right]$$

- A high flexible algo. can react strongly to random fluctuations on the training dataset. It results in Δ of fitted model as well. So flexibility reduces bias but inc. var.

- $$E\left[(Y - \hat{f}(x_0))^2\right] = \sigma^2 + \text{Bias}^2(\hat{f}(x_0)) + \text{Var}(\hat{f}(x_0))$$

Eg. $f(x_0) = 10$, $E[\hat{f}(x_0)] = 8$, $\text{Var}(\hat{f}(x_0)) = 3$, $\sigma^2 = 2$

$$\text{Bias} = 8 - 10 = -2$$
$$\text{MSE}(x_0) = 2 + (-2)^2 + 3 = 2 + 4 + 3 = 9$$

Eg.

	A	B
B	1	2
V	7	2
σ^2	3	3

$\text{MSE}_A = 1 + 7 + 3 = 11$
$\text{MSE}_B = 4 + 2 + 3 = 9$

- KNN bias & var.

$$\hat{f}(x_0) = \frac{1}{K} \sum_{i=1}^{K} y_i$$

$$E\left[\hat{f}(x_0) \mid X\right] = \frac{1}{K} \sum_{i=1}^{K} f(x_i)$$

$$\text{Bias}(x_0 \mid X) = \frac{1}{K} \sum_{i=1}^{K} f(x_i) - f(x_0)$$

$$\text{Var}\left(\hat{f}(x_0) \mid X\right) = \frac{\sigma^2}{K}$$

$$\text{MSE}(x_0 \mid X) = \sigma^2 + \frac{\sigma^2}{K} + \left[\frac{1}{K} \sum_{i=1}^{K} f(x_i) - f(x_0)\right]$$

Eg. $f(x_0) = 10, \quad \sigma^2 = 4, \qquad 9, 11, 13$

$K = 3$

$$E\left[\hat{f}(x_0) \mid X\right] = \frac{9+11+13}{3} = 11$$

$$\text{Bias} = 11 - 10 = 1 \qquad \text{Bias}^2 = 1$$

$$\text{MSE} = 1 + \frac{4}{3} + 4 = \frac{3+4+12}{3} = \frac{19}{3}$$

Eg. how to choose K, based on bias

$$\text{Bias}^2(K) = 0.01K^2, \quad \sigma^2 = 9$$

$$\text{MSE} = 9 + \frac{9}{K} + 0.01K^2$$

$$\frac{d}{dK}\text{MSE} = 0.02K - \frac{9}{K^2} = 0$$

$$0.02K^3 = 9 \Rightarrow K^3 = \frac{9}{0.02} \Rightarrow K = 7.66$$

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- when dim. increase

$$\lim_{p \to \infty} a^{1/p} = 1$$

$$V = l^p$$

$$l = a^{1/p}$$

$$a^{1/p} = \exp\left(\frac{\ln a}{p}\right) \qquad p \to \infty, \ \frac{\ln a}{p} \to 0$$

$$l \to e^0 = 1$$

- as dim. ↑, data becomes sparse relative to the enormous vol. of the input space. To collect enough neigh. for stable predic^n, local algo. has to search on a very large space.
- what amt. of data req. in high dim.

$$P(X \text{ belongs}) = h^p$$

$$E[k] = N h^p$$

$$\boxed{N \geq \frac{k}{h^p}}$$

- high dim. $p \Rightarrow$ nearby obs. diff. $\Rightarrow$ kNN neigh grow $\Rightarrow$ func^n values differ more $\Rightarrow$ sq. bias ↑ $\Rightarrow$ MSE ↑

$$\therefore \quad \boxed{|\text{Bias}(x_0)| \leq L R_k(x_0)}$$

or

$$\boxed{\text{Bias}^2(x_0) \leq L^2 R_k(x_0)^2}$$

- but linear models can escape this curse.

$$Y = X\beta + \varepsilon$$

$$\boxed{\text{Bias}(\hat{\beta} | X) = 0}$$

$$\boxed{MSE(x_0 | X) = \sigma^2 + \sigma^2 x_0^T (X^T X)^{-1} x_0}$$

$$\boxed{E[PE] = \sigma^2\left(1 + \frac{p}{N}\right)}$$

Eg. $N = 1000$, $p = 20$, $\sigma^2 = 4$

$$E[PE] = 4\left(1 + \frac{20}{1000}\right) = 4.08$$

$p = 100 \rightarrow E(PE) = 4.44$

- Smoothness → nearby inputs have similar outputs

$$|f(x) - f(z)| \leq L \|x - z\|$$

- Sparsity → only a small no. of features influence the response.

$$\|\beta_0\| = s, \quad s \leq p$$

- Additivity → $f(x) = \sum_{j=1}^{p} f_j(x_j)$
- Low-dim. space